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(54) **DRAIN FITTING AND INSTALLATION METHODS**

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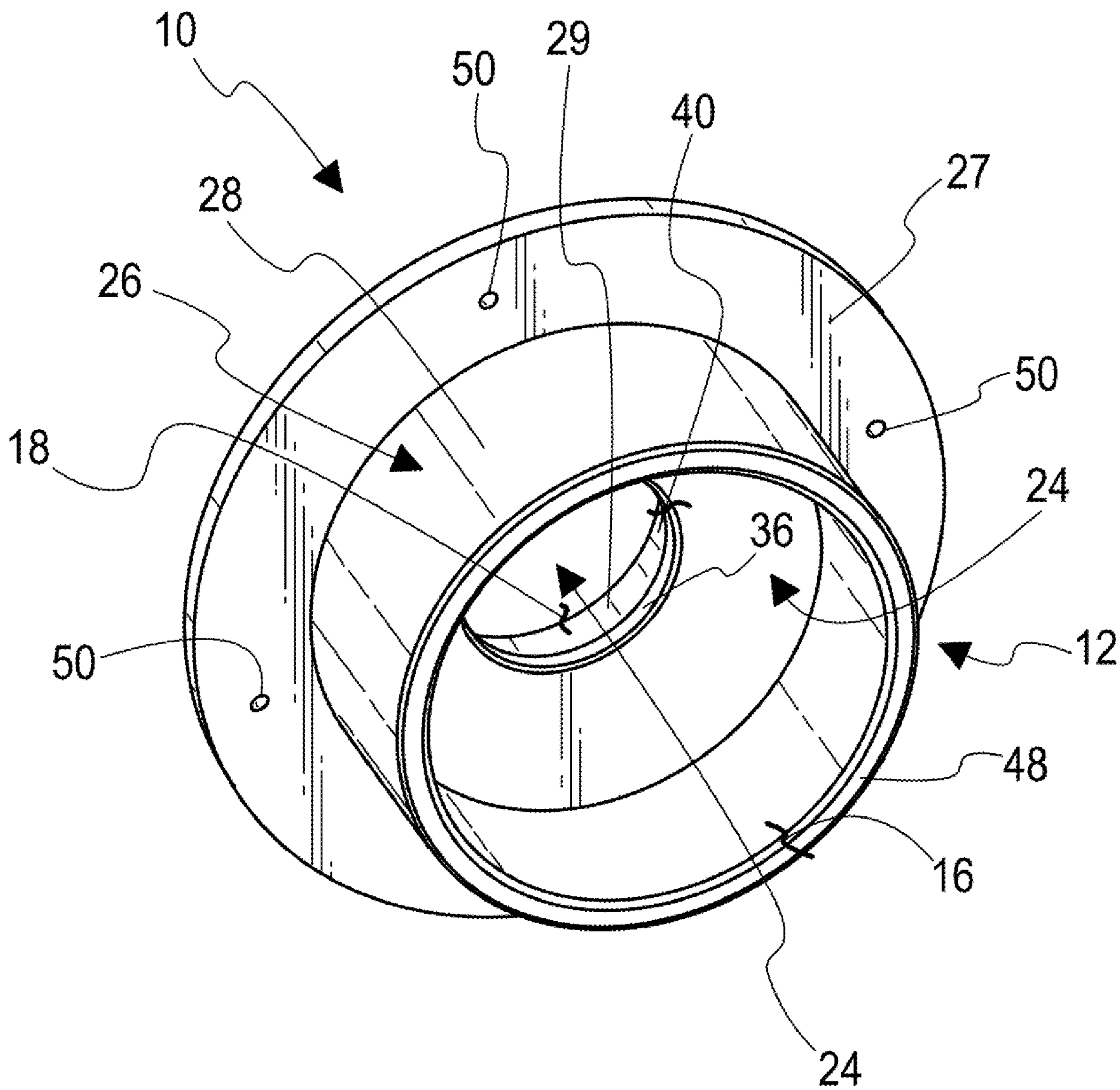
(57) **ABSTRACT**

A one-piece drain fitting with an integral mounting flange. The mounting flange has a plurality of holes for quickly securing the mounting flange to the top or bottom side of a wood sub-floor with screws or nails. The drain fitting can alternatively be installed in pre-cast or cast-in-place concrete without the use of fasteners. The drain fitting can be provided with a removable temporary cover for keeping out debris during construction.



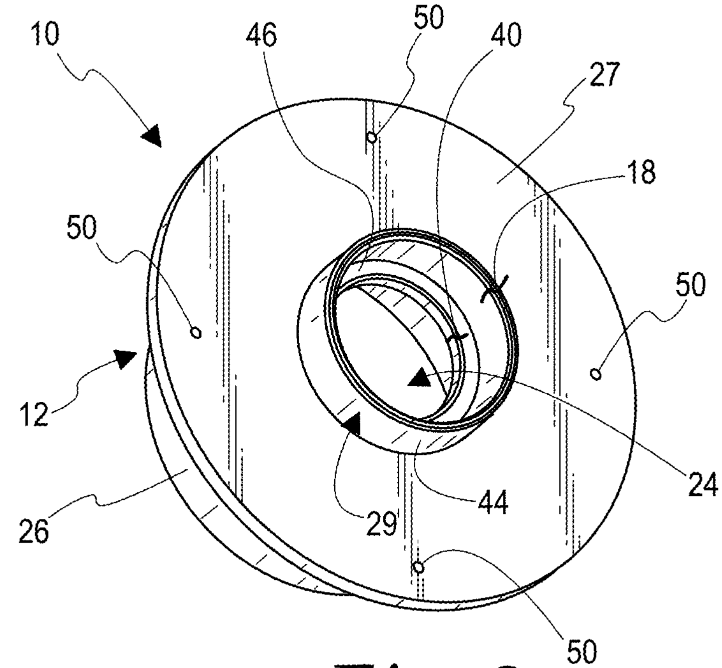

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Fig. 2

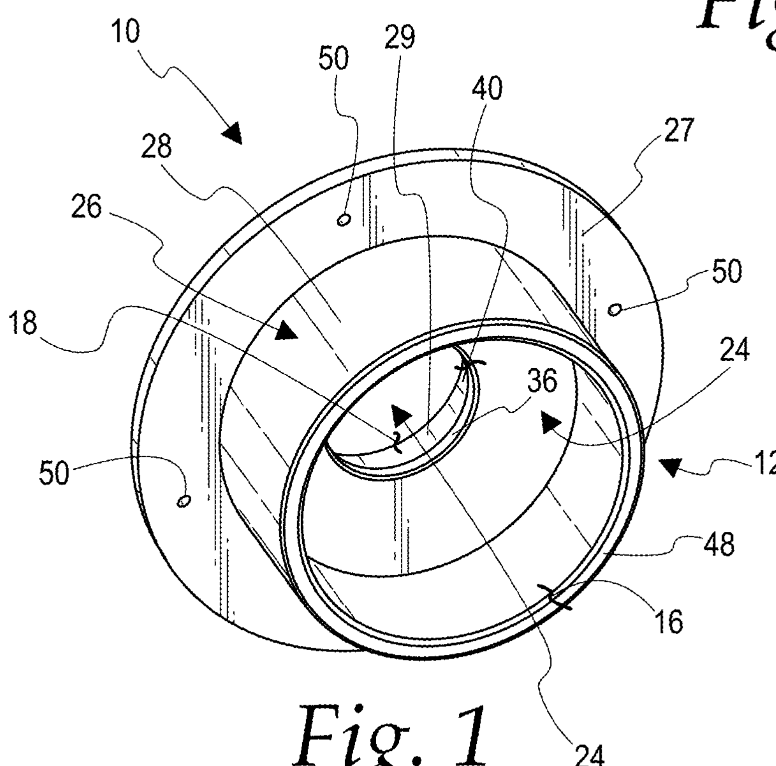
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Fig. 1

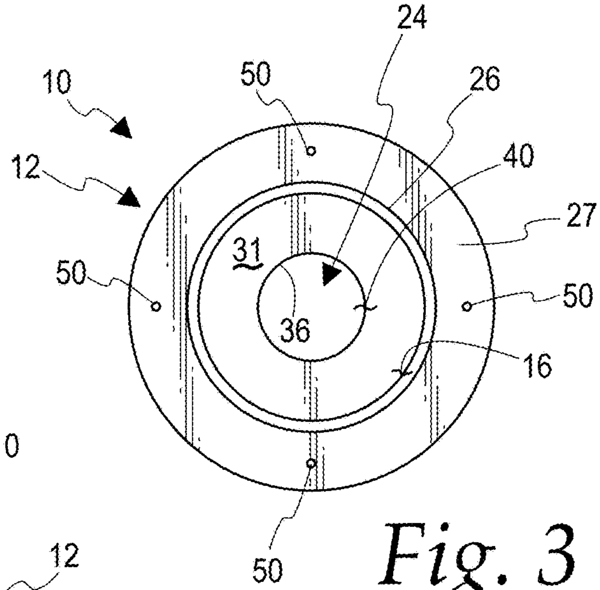

Fig. 3

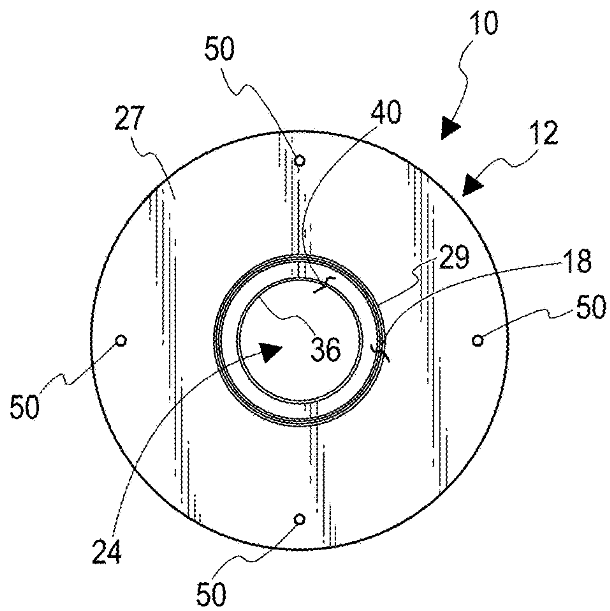

Fig. 4

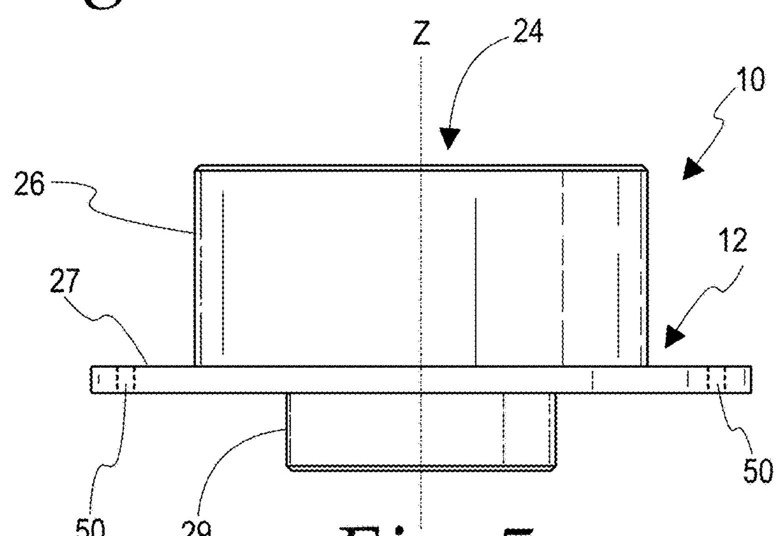

Fig. 5

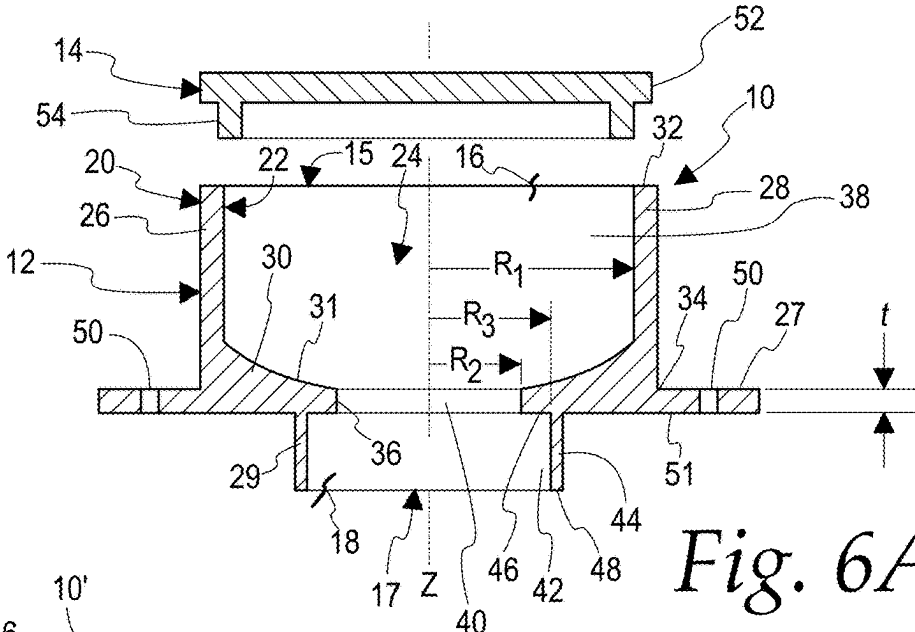

Fig. 6A

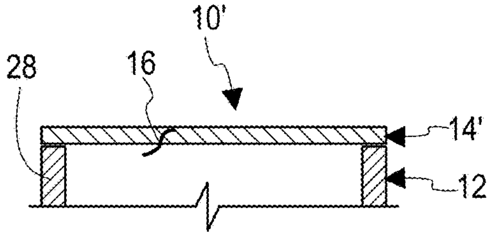

Fig. 6B

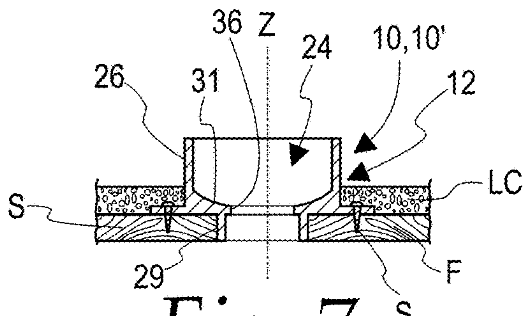

Fig. 7

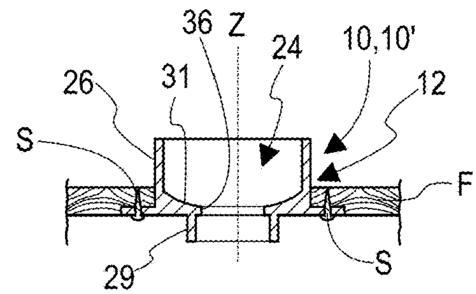

Fig. 8

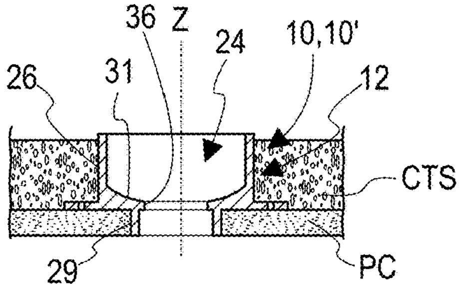

Fig. 9

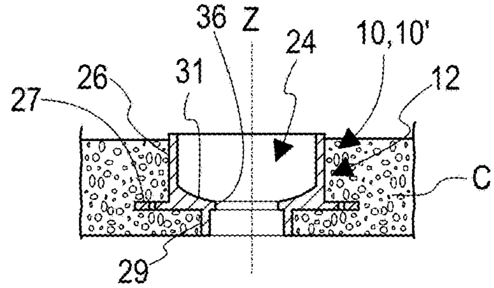

Fig. 10

DRAIN FITTING AND INSTALLATION METHODS

CROSS-REFERENCE

[0001] This application is a continuation of U.S. patent application Ser. No. 18/584,771, filed Feb. 22, 2024 and entitled DRAIN FITTING AND INSTALLATION METHODS, the entirety of which is hereby incorporated herein for all purposes.

FIELD OF THE DISCLOSURE

[0002] The present disclosure relates to a floor drain fitting. More particularly, it relates to a one-piece floor drain fitting for an open-site floor drain within a building and methods of installing the floor drain fitting in a floor.

SUMMARY

[0003] According to an aspect of the disclosure, a drain fitting comprises a drain body of one-piece construction. The drain body has a top side, a top opening on the top side, a bottom side opposite the top side, a bottom opening on the bottom side, an outer periphery, an inner periphery, an axis extending vertically through the top opening and the bottom opening, and a peripherally enclosed flow channel extending vertically from the top opening to the bottom opening and radially outwardly from the axis to the inner periphery. The drain body comprises a drain cup, a hub connection, and a peripheral mounting flange. The drain cup comprises a cup sidewall and a cup floor, the cup sidewall extending circumferentially around the flow channel and having an upper end circumscribing the top opening, the cup floor being connected to a lower end of the cup sidewall. The hub connection comprising a hub sidewall connected to a bottom side of the cup floor, the hub sidewall extending circumferentially around the flow channel, a lower end of the hub sidewall circumscribing the bottom opening. The mounting flange is connected to and extends radially outwardly from and circumferentially around the cup sidewall. The mounting flange comprises a plurality of fastener holes extending through a vertical thickness of the mounting flange, the mounting flange fastener holes being spaced radially outwardly from the cup sidewall and radially outwardly from the hub sidewall.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Although the characteristic features of this disclosure will be particularly pointed out in the claims, the disclosed method and system, and how it may be made and used, may be better understood by referring to the following description taken in connection with the accompanying drawings forming a part hereof, wherein like reference numerals refer to like parts throughout the several views and in which:

[0005] FIG. **1** is a top-lateral perspective view of a drain fitting according to an embodiment.

[0006] FIG. **2** is a bottom-lateral perspective view of the drain fitting of FIG. **1**.

[0007] FIG. **3** is a top plan view of the drain fitting of FIG. **1**.

[0008] FIG. **4** is a bottom plan view of the drain fitting of FIG. **1**.

[0009] FIG. **5** is a lateral elevation view of the drain fitting of FIG. **1**.

[0010] FIG. **6**A is a cross-sectional lateral elevation view of the drain fitting of FIG. **1** including a cover according to an embodiment.

[0011] FIG. **6**B is a truncated cross-sectional lateral elevation of the drain fitting of FIG. **1** including a cover according to another embodiment.

[0012] FIG. **7** is a cross-sectional elevation view of the drain fitting of FIG. **1** illustrating a method of installation according to an embodiment.

[0013] FIG. **8** is a cross-sectional elevation view of the drain fitting of FIG. **1** illustrating a method of installation according to another embodiment.

[0014] FIG. **9** is a cross-sectional elevation view of the drain fitting of FIG. **1** illustrating a method of installation according to another embodiment.

[0015] FIG. **10** is a cross-sectional elevation view of the drain fitting of FIG. **1** illustrating a method of installation according to another embodiment.

[0016] A person of ordinary skill in the art will appreciate that elements of the figures above are illustrated for simplicity and clarity and are not necessarily drawn to scale. The dimensions of some elements in the figures may have been exaggerated relative to other elements to help to understand the present teachings. Furthermore, a particular order in which certain elements, parts, components, modules, steps, actions, events and/or processes are described or illustrated may not be required. A person of ordinary skills in the art will appreciate that, for simplicity and clarity of illustration, some commonly known and well-understood elements that are useful and/or necessary in a commercially feasible embodiment may not be depicted to provide a clear view of various embodiments per the present teachings.

DETAILED DESCRIPTION

[0017] In the following description of various examples of embodiments of the disclosed device, reference is made to the accompanying drawings, which form a part hereof. Other specific arrangements of parts and environments can be used, and structural modifications and functional modifications can be made without departing from the scope of the disclosed device and methods.

[0018] As illustrated in the accompanying drawings and described herein, the present disclosure provides a one-piece floor drain fitting, including an integral mounting flange for fast and secure installation, for use as an open-site drain within a building.

[0019] Turning now to the drawings, shown in FIGS. **1**-**6**B are various views of a drain fitting **10**, **10'** according to embodiments of the disclosure. In one embodiment, the drain fitting **10** comprises a drain body **12** of one-piece construction and a cover **14** that is a simple insertable plug-type cover. In another embodiment, the drain fitting **10'** comprises the drain body **12** and a cover **14'** that is a thin, breakable film affixed to an upper end of the drain body **12**, such as by heat welding or the application of a suitable adhesive. The drain body **12** can be formed, for example, in one piece of suitable polymer such as molded PVC or molded CPVC, cast iron, or other suitable plumbing material. As best seen in the lateral cross-sectional elevation view of FIG. **6**A, the drain body **12** has a top side **15**, a top opening **16** on the top side **15**, a bottom side **17** opposite the top side **15**, a bottom opening **18** on the bottom side **17**, an outer periphery **20**, an inner periphery **22**, a z-axis extending vertically through the top opening **16** and the bottom opening **18**, and a peripherally enclosed flow channel **24** extending vertically from the top opening **16** to the bottom opening **18** and radially outwardly from the z-axis to the inner periphery **22**.

[0020] The drain body **12** comprises a drain cup **26**, a peripheral mounting flange **27**, and a hub connection **29**. The drain cup **26** comprises a cup sidewall **28** and a cup floor **30**. The cup sidewall **28** extends circumferentially around the flow channel **24** and has an upper end **32** circumscribing the top opening **16**, the cup floor **30** being connected to a lower end **34** of the cup sidewall **28**. The cup floor **30** has an upper surface **31**, the cup floor upper surface **31** extending circumferentially around an upper end of a central opening formed in the cup floor **30**, and radially outwardly and vertically upwardly from the central opening to the cup sidewall **28**. An average downward slope in a radially inward direction from an upper end to a lower end of the cup floor upper surface **31** can be, for example, in a range from 1:3 to 2:3. More particularly, the cup floor upper surface **31** is upwardly concave and serves to direct drainage toward a throat **36** that forms the central opening and extends downwardly through the cup floor **30**. In other embodiments not shown, that slope can be continuous so that the cup floor upper surface is conical/frustoconical. The throat **36** thus fluidly connects an interior volume of the drain cup **26** below the top opening **16** to an interior volume of the hub connection **29** above the bottom opening **18**.

[0021] The hub connection **29** comprises a hub sidewall **44** connected to a lower side **46** of the cup floor **30**, the lower side **46** comprising a lower surface of the cup floor **30**, which circumscribes a lower end of the throat **36** and extends radially outwardly from the throat **36** to the hub sidewall **44**. The lower side **46** of the cup floor **30** serves as a stop for receiving an upper end of a drain pipe inserted into the hub connection **29**. More particularly, an inner diameter of the throat **36** corresponds to an inner diameter of a drain pipe to be so inserted. The hub sidewall **44** extends circumferentially around the flow channel **24**, a lower end **48** of the hub sidewall **44** circumscribing the bottom opening **18**.

[0022] The mounting flange **27** is connected to and extends radially outwardly from and circumferentially around the cup sidewall **28**. The mounting flange comprises a plurality of fastener holes **50** extending through a vertical thickness t of the mounting flange, which may for example be ⅛ inch to ⅜ inch, or more particularly ¼ inch. The mounting flange fastener holes **50** are spaced radially outwardly from the cup sidewall **28** and radially outwardly from the hub sidewall **44**. As illustrated, the mounting flange **27** has the shape of an annular disk. More particularly, the mounting flange **27** comprises a planar lower surface **51**, the mounting flange lower surface **51** being coplanar with that of the lower side **46** of the cup floor **30**.

[0023] Accordingly, the flow channel **24** comprises a cup region **38**, a throat region **40**, and a hub region **42**, the cup region **38** extending downwardly from the top opening **16** to the upper surface **31** of the cup floor **30** and the throat region **40**, the throat region **40** extending downwardly from the cup region **38** to the hub region **42**, the hub region **42** extending downwardly from the throat region **40**, and from the lower side **46** of the cup floor **30**, to the bottom opening **18**. The flow channel cup region **38** extends to a first radius R_1 from the z-axis, the flow channel throat region **40** extending to a second radius R_2 from the z-axis, and the flow channel hub region **42** extends to a third radius R_3 from the z-axis. The second radius R_2 is smaller than the first radius R_1 and smaller than the third radius R_3. The drain cup **26** can have a pipe size (i.e., a nominal inner diameter), for example, ranging from 1-½ inches to 6 inches. More particularly, the drain cup **26** can have a 3-inch or 4-inch pipe size, and can comprise a length of 3-inch or 4-inch Schedule 40 PVC pipe. The hub connection **29** can have a hub size (i.e., can fit a pipe of nominal inner diameter), for example, ranging from 1-¼ inches to 3 inches. More particularly, the hub connection **29** can have a hub size of 1-½ inches or 2 inches, and still more particularly can be a 1-½-inch or 2-inch Schedule 40 PVC hub, with a wall thickness of about 4 mm, corresponding to an inner diameter of the hub connection **29** of 1.900 inch or 2.375 inches, respectively.

[0024] As mentioned above, in embodiments of the disclosure, the drain fitting **10**, **10'** further comprises a cover **14**, **14'**. As shown in FIGS. **6**A and **6**B, the cover **14**, **14'** is removably disposed over the upper end **32** of the cup sidewall **28** so as to cover the top opening **16** of the drain body **12**. The cover **14**, **14'** serves to keep debris out of the drain fitting **10**, **10'**, which can be used as an open-site drain within a building, during construction, and can then be removed. As illustrated in FIG. **6**A, the cover **14** is a one-piece rigid body comprising a flat round disk **52** and a downward projection **54** connected to the bottom side of the disk **52**, the downward projection **54** being operative to be removably inserted into the top opening **16** of the drain body **12** so that an outer periphery of the disk **52** overlaps an inner periphery of the upper end **32** of the cup sidewall **28**. The disk **52** can thus simply rest on top of the upper end **32** of the cup sidewall **28**, so that the cover **14** is removable simply by being lifted manually from the drain body **12**. With reference to FIG. **6**B, the film of the cover **14'**, which may for example be about 1/64 inch to about ⅛ inch thick, is affixed to the upper end **32** of the cup sidewall **28**, such as by heat welding or a suitable adhesive as mentioned above, in such a manner that it can be manually peeled away. In an embodiment, such peeling can be facilitated by the cover **14'** including a manually graspable tab (not shown) that extends outside the outer periphery of the cup sidewall **28**. Alternatively, the film of the cover **14'** can be punctured by a hand tool, such as a utility knife or screwdriver, to form a hole, to permit a user to tear the film from the hole to its outer edge and then grasp and peel away the outer edge of the film from the cup sidewall upper end **32**.

[0025] Four methods of installing the floor drain fitting **10**, **10'** are illustrated in FIGS. **7**-**10**, respectively, where the cover **14**, **14'** is omitted (thus illustrating the drain fitting **10**, **10'** with the cover **14**, **14'** already removed after construction is completed) for ease of illustration. With reference to FIGS. **7** and **8**, two methods are illustrated in which the mounting flange **27** is fastened by fasteners (illustrated as screws S) to a subfloor F, where the subfloor F can be made of wood or other suitable material for driving screws or nails. According to each method, the drain body **12** is positioned so that the mounting flange **27** is disposed against a surface of the subfloor F and so that the subfloor surface overlaps the fastener holes **50**, and the fasteners (screws S) are driven vertically through the fastener holes **50** and into the subfloor F to fasten the mounting flange **27** to the subfloor F. In the method shown in FIG. **7**, the mounting flange **27** is placed over the subfloor F so that the hub connection **29** extends downwardly through a hole formed in the subfloor F, the subfloor surface that overlaps the fastener holes **50** being an upper surface of the subfloor F, and the screws S are driven downwardly. A layer of gypsum concrete can then be formed or placed over the subfloor F and the mounting flange **27** of the drain fitting **10**, **10'**. In the method shown in FIG. **8**, the drain body **12** is positioned so that the drain cup **26** extends upwardly through a hole formed in the subfloor F, the subfloor surface that overlaps the fastener holes **50** being a lower surface of the subfloor F, and the screws S are driven upwardly (that is, upwardly with respect to the installed orientation of the subfloor F).

[0026] Turning to FIGS. **9** and **10**, two more methods are illustrated of using the drain fitting **10**, **10'** in concrete-only floor installations. As illustrated in FIG. **9**, the drain body **12** is positioned so that the hub connection **29** is inserted downwardly into a countersunk upper end of a hole in a precast concrete form PC, and the mounting flange **27** rests on top of the precast concrete form PC. A concrete topping slab CTS is then poured over the drain fitting **10**, **10'**. According to a fourth method shown in FIG. **10**, a cast-in-place concrete slab is poured around the drain fitting **10**, **10'**, such as by sliding the hub connection **29** onto the upper end of a drain pipe (not shown) sized to be received in the hub connection **29** until the upper end of the drain pipe meets the lower side **46**, and then pouring concrete to fill the peripheral space around the drain pipe and the drain fitting **10**, **10'** up to a level above the mounting flange **27** and below the upper end of the drain cup **26**. The mounting flange **27**, while not attached to the concrete by fasteners in either of these methods, does not interfere with the installation process, while allowing the drain fitting **10**, **10'** to be held down in place by the concrete disposed over the mounting flange **27** in the installed position.

[0027] The preceding description of the disclosure has been presented for purposes of illustration and description and is not intended to be exhaustive or to limit the disclosure to the precise form disclosed. The description was selected to best explain the principles of the present teachings and practical application of these principles to enable others skilled in the art to best utilize the disclosure in various embodiments and various modifications as are suited to the particular use contemplated. It should be recognized that the words “a” or “an” are intended to include both the singular and the plural. Conversely, any reference to plural elements shall, where appropriate, include the singular.

[0028] It is intended that the scope of the disclosure not be limited by the specification, but be defined by the claims set forth below. In addition, although narrow claims may be presented below, it should be recognized that the scope of this disclosure is much broader than presented by the claim(s). It is intended that broader claims will be submitted in one or more applications that claim the benefit of priority from this application. Insofar as the description above and the accompanying drawings disclose additional subject matter that is not within the scope of the claim or claims below, the additional disclosures are not dedicated to the public and the right to file one or more applications to claim such additional disclosures is reserved.

1. (canceled)

2. The drain fitting of claim **14** wherein the drain body is formed of molded PVC.

3. The drain fitting of claim **14**, further comprising a cover, the cover being removably disposed over an upper end of the cup sidewall so as to cover the top opening.

4. A drain fitting comprising:

a drain body of one-piece construction, the drain body having a top side, a top opening on the top side, a bottom side opposite the top side, a bottom opening on the bottom side, an outer periphery, an inner periphery, an axis extending vertically through the top opening and the bottom opening, and a peripherally enclosed flow channel extending vertically from the top opening to the bottom opening and radially outwardly from the axis to the inner periphery;

the drain body comprising:

a drain cup, the drain cup comprising a cup sidewall and a cup floor, the cup sidewall extending circumferentially around the flow channel and having an upper end circumscribing the top opening, the cup floor being connected to a lower end of the cup sidewall, the flow channel comprising a throat region extending vertically through the cup floor;

a hub connection, the hub connection comprising a hub sidewall connected to a bottom side of the cup floor, the hub sidewall extending circumferentially around the flow channel, a lower end of the hub sidewall circumscribing the bottom opening;

a peripheral mounting flange, the mounting flange being connected to and extending radially outwardly from and circumferentially around the cup sidewall, the mounting flange comprising at least one fastener hole extending through a vertical thickness of the mounting flange, the mounting flange fastener hole being spaced radially outwardly from the cup sidewall and radially outwardly from the hub sidewall;

a cover, the cover being removably disposed over an upper end of the cup sidewall so as to cover the top opening, the cover comprising a breakable film affixed to the upper end of the cup sidewall.

5. The drain fitting of claim **4** wherein the breakable film is affixed by one of welding and a suitable adhesive being applied between the breakable film and the upper end of the cup sidewall.

6. The drain fitting of claim **3** wherein the cover is a one-piece rigid body comprising a flat round disk and a downward projection connected to the bottom side of the disk, the downward projection being operative to be removably inserted into the top opening of the drain body so that an outer periphery of the disk overlaps an inner periphery of the upper end of the cup sidewall.

7. The drain fitting of claim **14** wherein the flow channel further comprises a cup region and a hub region, the cup region extending downwardly from the top opening to the cup floor and the throat region, the throat region extending downwardly from the cup region to the hub region, the hub region extending downwardly from the throat region to the bottom opening, the flow channel cup region extending to a first radius from the axis, the flow channel throat region extending to a second radius from the axis, and the flow channel hub region extending to a third radius from the axis, the second radius being smaller than the first radius and smaller than the third radius.

8. The drain fitting of claim **7** wherein the third radius is smaller than the first radius.

9. The drain fitting of claim **8** wherein the flow channel cup region comprises a cylindrical volume having a first diameter that is twice the first radius, the flow channel throat region comprises a cylindrical volume having a second diameter that is twice the second radius, and the flow channel hub region comprises a cylindrical volume having a third diameter that is twice the third radius.

10. The drain fitting of claim **9**, further comprising the first diameter measuring in a range from 2 ½ inches to 5 inches and the third diameter measuring in a range from 1 ½ inch to 3 inches.

11. The drain fitting of claim **9** wherein the first diameter has a measurement selected from **3** inches and **4** inches and the second diameter has a measurement selected from 1.900 inch and 2.375 inches.

12. The drain fitting of claim **14** wherein the cup floor comprises an upper surface extending circumferentially around an upper end of the flow channel throat region and radially outwardly and vertically upwardly from the flow channel throat region to the cup sidewall.

13. The drain fitting of claim **12** wherein the cup floor upper surface is upwardly concave.

14. A drain fitting comprising:

a drain body of one-piece construction, the drain body having a top side, a top opening on the top side, a bottom side opposite the top side, a bottom opening on the bottom side, an outer periphery, an inner periphery, an axis extending vertically through the top opening and the bottom opening, and a peripherally enclosed flow channel extending vertically from the top opening to the bottom opening and radially outwardly from the axis to the inner periphery;

the drain body comprising:

a drain cup, the drain cup comprising a cup sidewall and a cup floor, the cup sidewall extending circumferentially around the flow channel and having an upper end circumscribing the top opening, the cup floor being connected to a lower end of the cup sidewall, the flow channel comprising a throat region extending vertically through the cup floor;

a hub connection, the hub connection comprising a hub sidewall connected to a bottom side of the cup floor, the hub sidewall extending circumferentially around the flow channel, a lower end of the hub sidewall circumscribing the bottom opening;

a peripheral mounting flange, the mounting flange being connected to and extending radially outwardly from and circumferentially around the cup sidewall, the mounting flange comprising at least one fastener hole extending through a vertical thickness of the mounting flange, the mounting flange fastener hole being spaced radially outwardly from the cup sidewall and radially outwardly from the hub sidewall;

each of the cup floor and the mounting flange having a planar lower surface, the cup floor lower surface being coplanar with the mounting flange lower surface.

15. The drain fitting of claim **14** wherein the cup floor lower surface circumscribes a lower end of the flow channel throat region and extends radially outwardly from the flow channel throat region to the hub sidewall.

16. (canceled)

17. A method of installing a floor drain fitting, the floor drain fitting comprising a one-piece drain body having a top side, a top opening on the top side, a bottom side opposite the top side, a bottom opening on the bottom side, an outer periphery, an inner periphery, an axis extending vertically through the top opening and the bottom opening, and a peripherally enclosed flow channel extending vertically from the top opening to the bottom opening and radially outwardly from the axis to the inner periphery; the drain body comprising a drain cup, a hub connection, and a peripheral mounting flange, the drain cup circumscribing the top opening, the hub connection circumscribing the bottom opening, the mounting flange being connected to and extending radially outwardly from and circumferentially around the drain cup, the mounting flange comprising at least one fastener hole extending through a vertical thickness of the mounting flange, the mounting flange fastener hole being spaced radially outwardly from the drain cup and the hub connection, the method comprising:

positioning the drain body so that the drain cup extends upwardly through a hole formed in a subfloor, so that the mounting flange is disposed against a lower surface of the subfloor, and so that the subfloor lower surface overlaps the fastener holes; and

driving the fasteners upwardly through the fastener holes and into the subfloor to fasten the mounting flange to the subfloor.

18. A method of installing a floor drain fitting, the floor drain fitting comprising a one-piece drain body having a top side, a top opening on the top side, a bottom side opposite the top side, a bottom opening on the bottom side, an outer periphery, an inner periphery, an axis extending vertically through the top opening and the bottom opening, and a peripherally enclosed flow channel extending vertically from the top opening to the bottom opening and radially outwardly from the axis to the inner periphery; the drain body comprising a drain cup, a hub connection, and a peripheral mounting flange, the drain cup circumscribing the top opening, the hub connection circumscribing the bottom opening, the mounting flange being connected to and extending radially outwardly from and circumferentially around the drain cup, the mounting flange comprising at least one fastener hole extending through a vertical thickness of the mounting flange, the mounting flange fastener hole being spaced radially outwardly from the drain cup and the hub connection, the method comprising:

positioning the drain body so that the hub connection extends downwardly through a hole formed in a subfloor, so that the mounting flange is disposed against an upper surface of the subfloor, and so that the upper subfloor surface overlaps the fastener holes, and driving the fasteners downwardly through the fastener holes and into the subfloor to fasten the mounting flange to the subfloor.

19. The drain fitting of claim **4** wherein the flow channel further comprises a cup region and a hub region, the cup region extending downwardly from the top opening to the cup floor and the throat region, the throat region extending downwardly from the cup region to the hub region, the hub region extending downwardly from the throat region to the bottom opening, the flow channel cup region extending to a first radius from the axis, the flow channel throat region extending to a second radius from the axis, and the flow channel hub region extending to a third radius from the axis, the second radius being smaller than the first radius and smaller than the third radius.

20. The drain fitting of claim **19** wherein the third radius is smaller than the first radius.

21. The drain fitting of claim **20** wherein the flow channel cup region comprises a cylindrical volume having a first diameter that is twice the first radius, the flow channel throat region comprises a cylindrical volume having a second diameter that is twice the second radius, and the flow channel hub region comprises a cylindrical volume having a third diameter that is twice the third radius.

22. The drain fitting of claim **21**, further comprising the first diameter measuring in a range from 2 ½ inches to 5 inches and the third diameter measuring in a range from 1 ½ inch to 3 inches.

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